

SPARSE: a sparse hypergraph neural network for learning multiple types of latent combinations to accurately predict drug-drug interactions

Outline

- Background
- Proposed method
- Experiments
- Conclusion

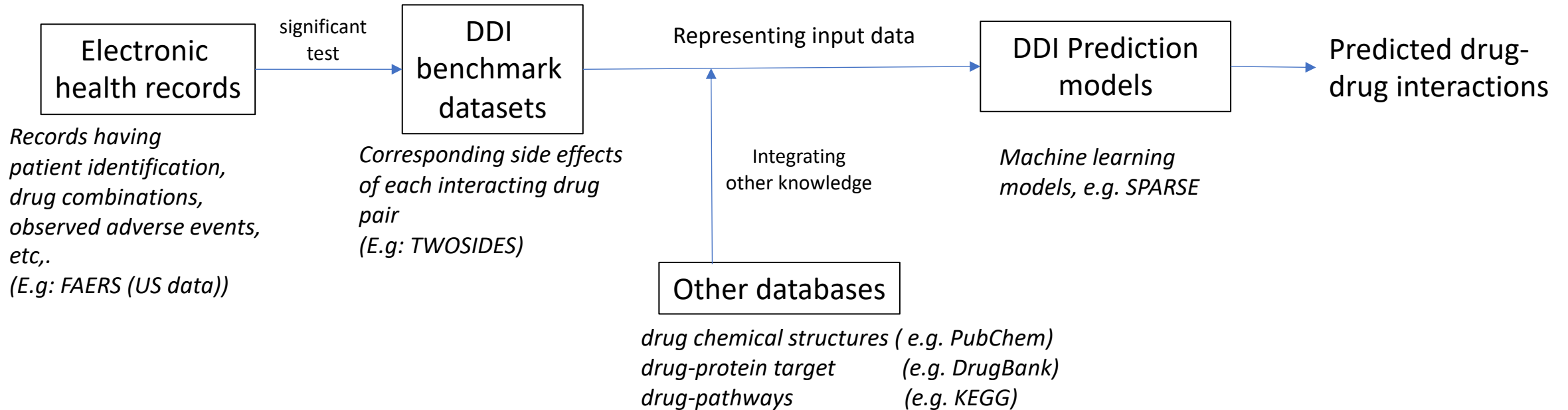
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Terminologies

- **An adverse event** is any undesirable experience associated with the use of a medical product in a patient.
- **Side effects**, also known as **adverse reactions**, are unwanted undesirable effects that are possibly related to a drug.
- **A drug-drug interaction (DDI)** is a reaction between two drugs, whereby the effects of one drug are modified by the concomitant use of the second drug.
- **Graph**: a data structure with nodes and edges, where each edge consists of two nodes.
- **Hypergraph**: a generalization of the graph with nodes and hyperedges, where each hyperedge might consists of *more than* two nodes.

Data flow for constructing drug-drug interaction (DDI) prediction models



Electronic health records

significant test

patient id	Drug combinations	Adverse event (ae)
patient_1	drug_1, drug_2, drug_3	ae_1, ae_2
patient_2	drug_2, drug_3	ae_2, ae_3
patient_3	drug_1, drug_3	ae_1, ae_4
patient_4	drug_3	ae_4
...	...	...

DDI benchmark
datasets

Drug pair	Adverse events
drug_2, drug_3	ae_2
drug_1, drug_3	ae_1, ae_5
...	...

Significant test for (drug_i, drug_j) and ae_z

Commonly used: Fisher’s extract test

To check whether the chance of occurring ae_z in cases having (drug_i, drug_j) is different (higher) than the chance of occurring ae_z in cases not having (drug_i, drug_j)

Contingency table

		(drug_i, drug_j)	Other drug combinations
ae_z	Yes	Count_1	Count_3
	No	Count_2	Count_4

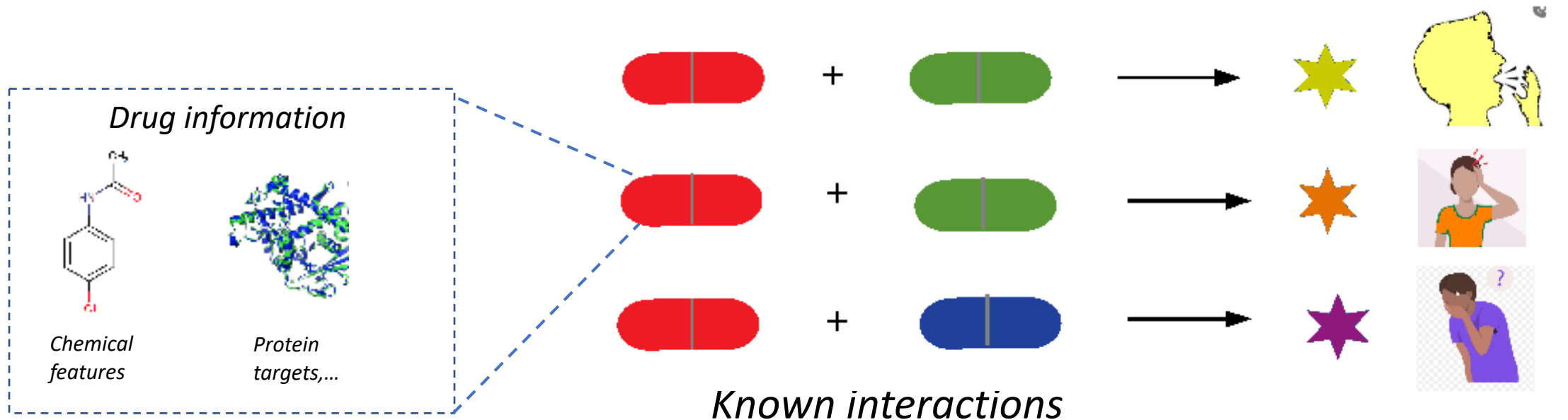
>?

Notes on extracting DDI benchmark datasets:

- The significant test can only extract association relationships, not casual relationship. If an adverse event is highly associated with a drug pair, it does not mean that the drug pair causes the adverse event.
- Some studies consider Adverse events as Side effects.
- Some important information, e.g. amount of drugs, time of using drugs, route of drug administration, has not been considered.
- The current commonly used DDI benchmark dataset (TWOSIDES) are actually the datasets for highly associated adverse events with each drug pair. (But there is no better ones with hundreds thousands of DDIs.)

Drug-Drug Interaction (DDI) Prediction Task

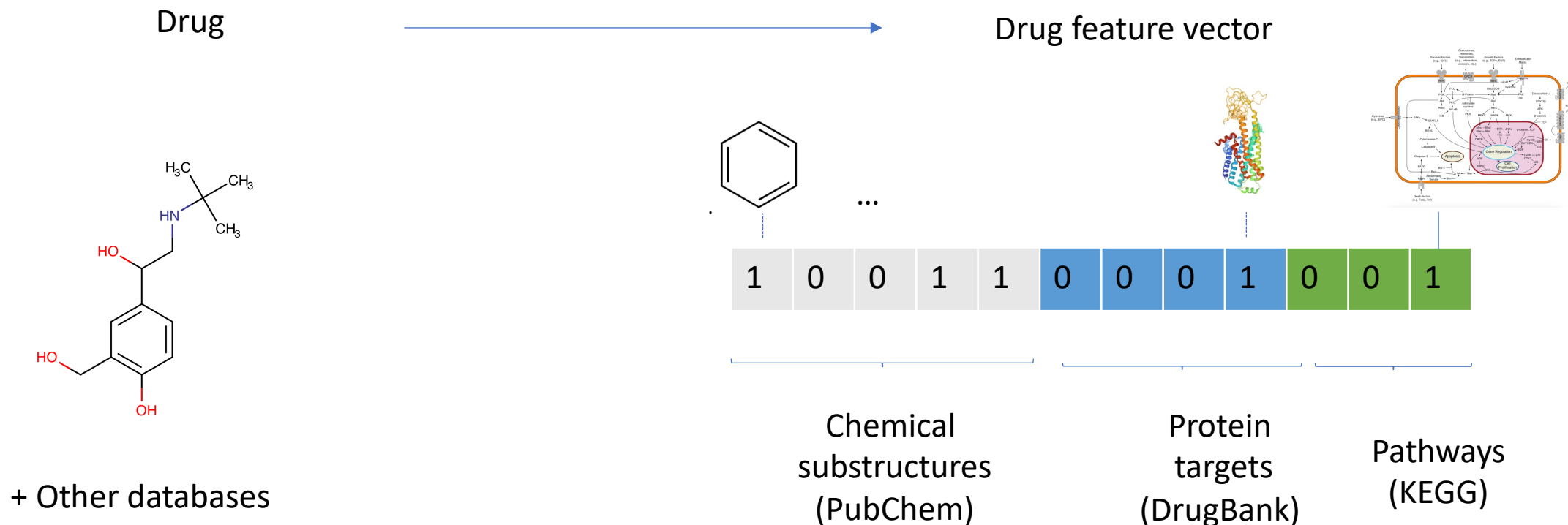
DDI Data



Predict all side effects for each drug pair

Integrating other databases to represent drug information

Each drug can be represented by a binary vector (drug feature vector) indicating the presences of its chemical substructures, associated proteins, and pathways.

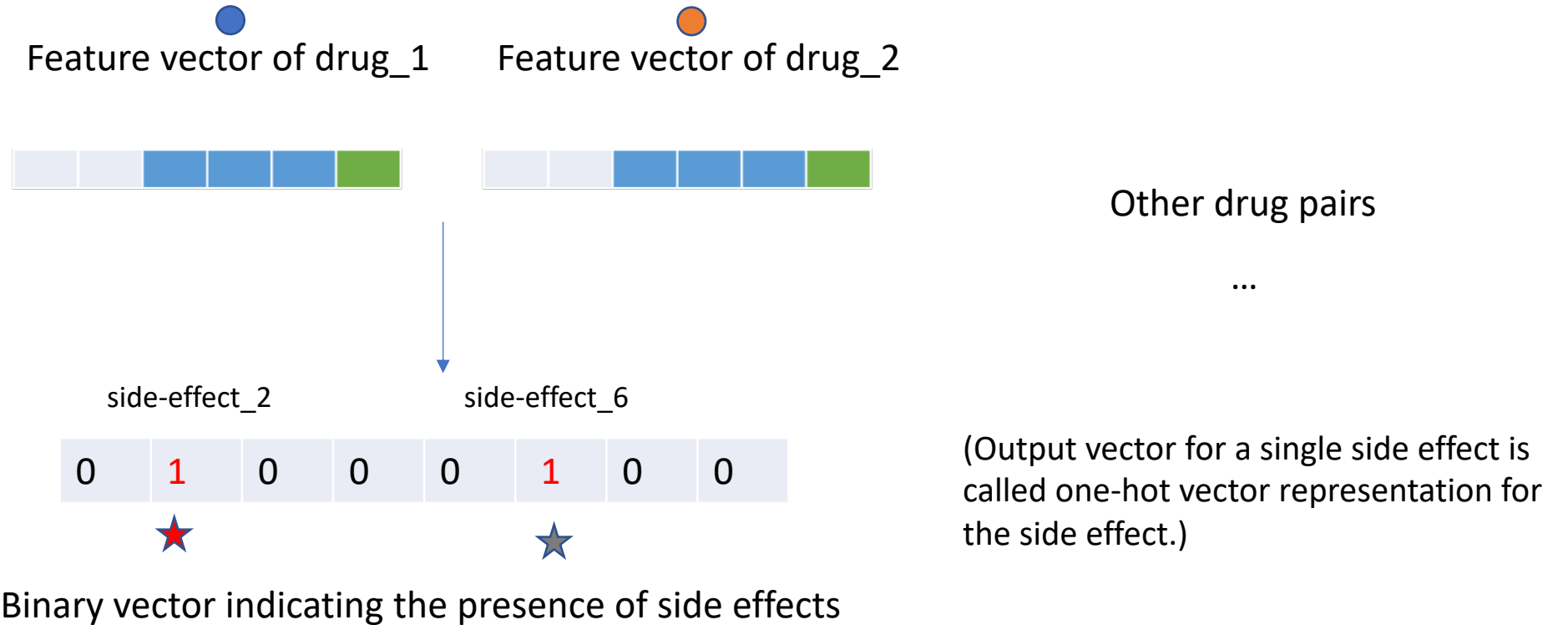


(Note: In practice, the size of the drug feature vector might up to thousands.)

Representing DDI data

Represent DDIs data in the forms of commonly used data structures (vectors, graphs, etc) for mathematical models.

1) Pair of drug feature vectors and a binary vector indicating the corresponding side effects



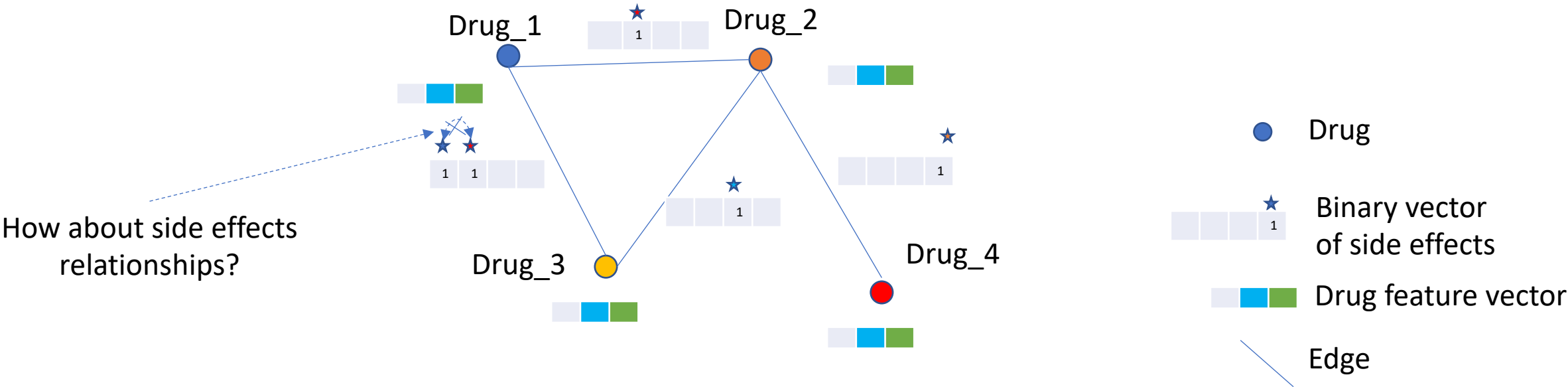
This representation is simple, which can be used for algorithms taking input as a pair of vectors (or a single vector after concatenating the two input vectors) and output as a vector.

But this representation is not effective to represent sharing drugs between different drug pairs (Propagating information among many drugs?).

2) Use graph data to represent all interacting drug-pairs

Node: drug with feature vector

Edge: Interacting drug pair with labels as a binary vector indicating corresponding side effects



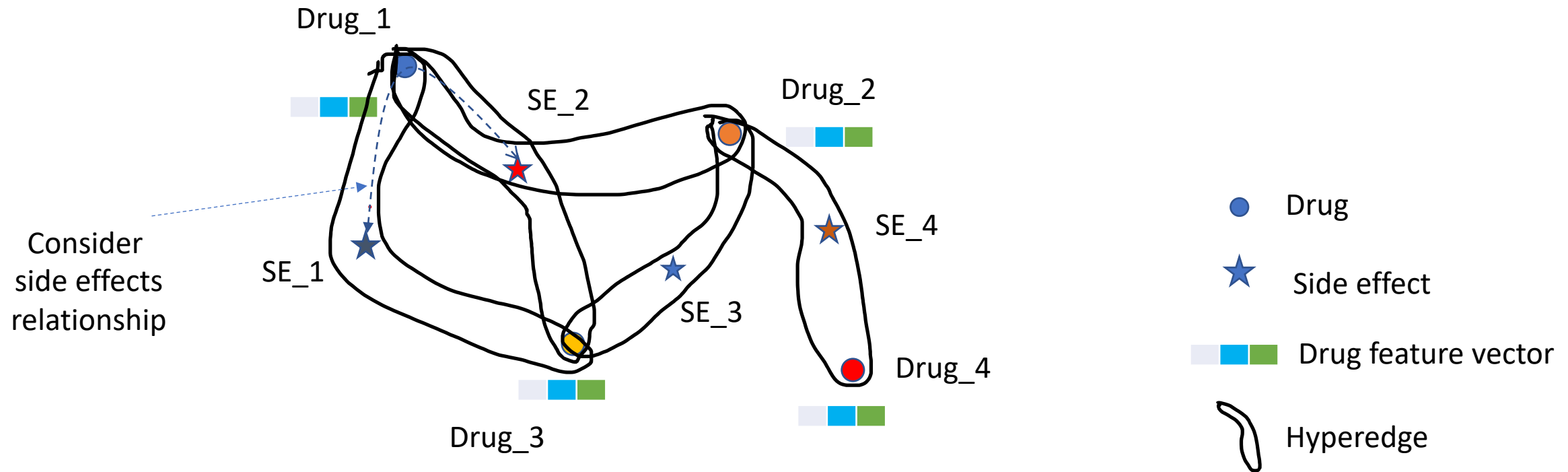
Propagating drug information through edges. Can use learning methods on graphs.

How to propagating drugs and side effects altogether?

3) Use hypergraph data to represent DDIs

Node : Drug (with feature vector) or Side effect (with one-hot feature vector)

Hyperedge: Triple of two interacting drug and a corresponding side effect. (Each drug pair might have more than one hyperedge, each hyperedge corresponds to a side effect of the drug pair.)



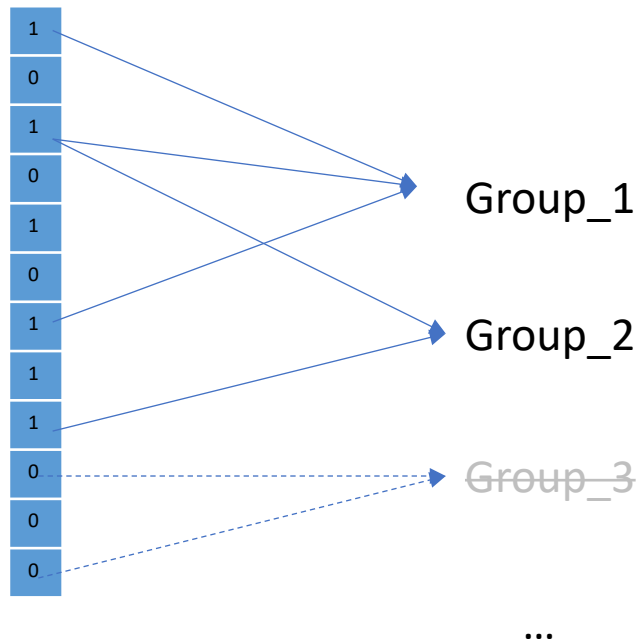
A side effect might have feature vector describing side effect properties (classes).

Learning on the hypergraph to propagate information of drugs and side effects altogether.

Latent groups (latent features)

Assuming that there exist some groups of input features (chemical substructures, proteins, pathways) forming some functions of drugs, which also can take responsibility to cause side effects.

Observed features Latent groups (latent features)



These groups are normally not directly observed, called latent groups (latent features).

Instead of using original features, we can use the presence of the latent features to describe (function of) drugs.

The latent features can be more complicated.
E.g. In hierarchical structures, non-linearity combinations.

Problem:

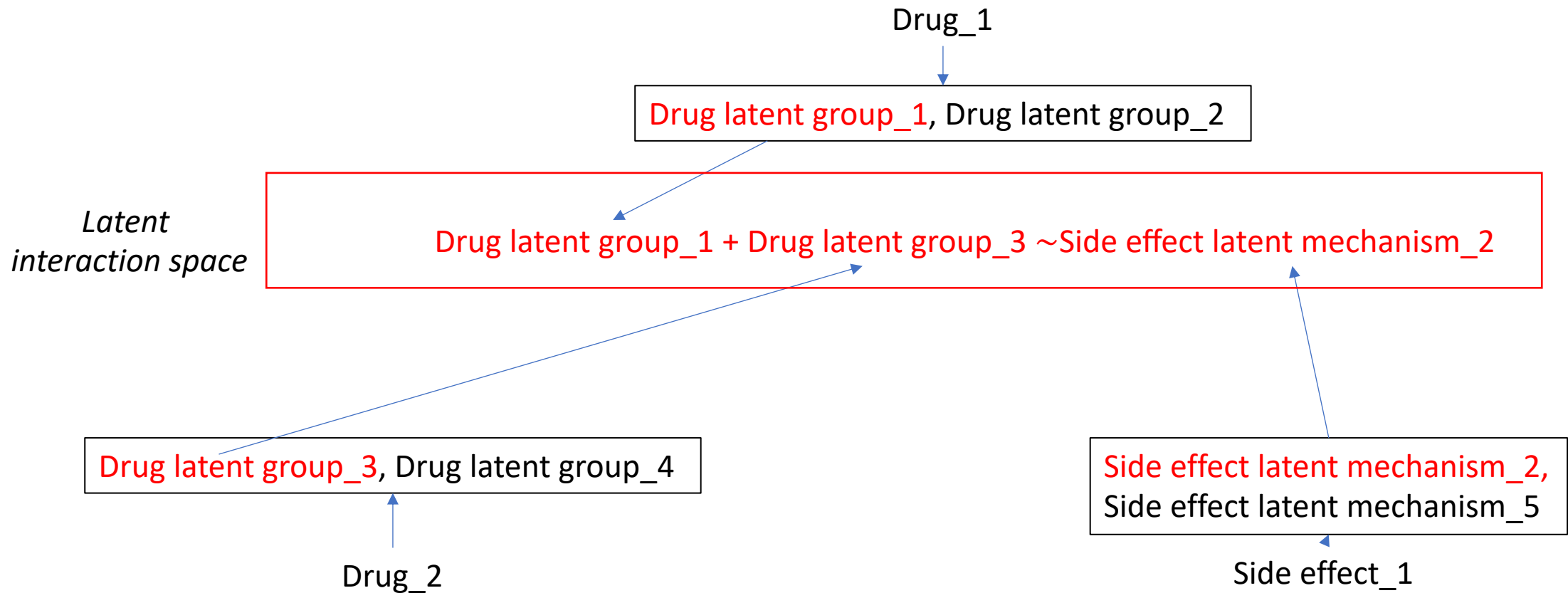
How to find these latent features?

What is the role of the latent features in causing side effects?

Outline

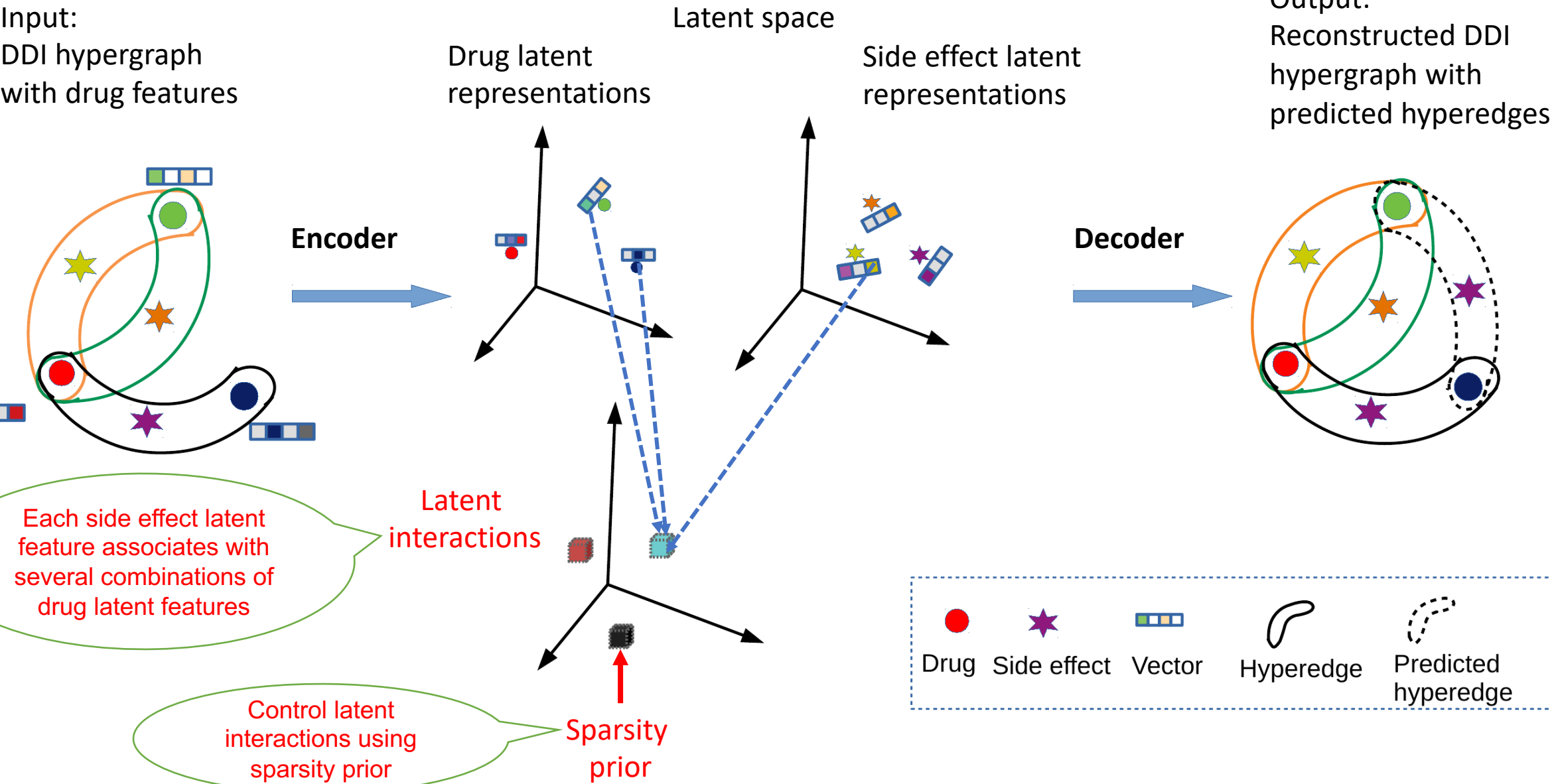
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We proposed a new model (SPARSE) that can learn latent groups of drugs and side effects with their interactions.

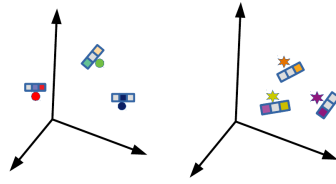


A simple illustration for the latent interaction assumption of drug-drug-side effect interactions.

SPARSE: Full illustration



SPARSE: Computation



$$H \in (R^{|V_D| \times K_D}, R^{|V_S| \times K_S})$$



$(\tilde{G})$

$f_1(G, F)$

$f_2(G, F)$

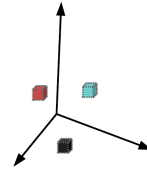
(G, F)

$G = (V, E)$

$$V = V_D \cup V_S$$

$$E \subset V_D \times V_D \times V_S$$

$$F \in (R^{|V_D| \times C}, R^{|V_S| \times |V_S|})$$



$$B \in R^{K_D \times K_D \times K_S}$$

$Prior(\Lambda)$

SPARSE_O : No prior

SPARSE_L : Laplace prior

SPARSE : Horseshoe prior

$$\tilde{e} \sim N(m(\tilde{e}), \delta)$$

$$\tilde{e} = (u, v, t) \in V_D \times V_D \times V_S$$

$$m(\tilde{e}) = B \times_1 h_u \times_2 h_v \times_3 h_t$$

DDI score

$$B_{i,j,k} \sim N(0, \tau^2 \Lambda_{i,j,k}^2)$$

$$\Lambda_{i,j,k} \sim C(0,1)^+$$

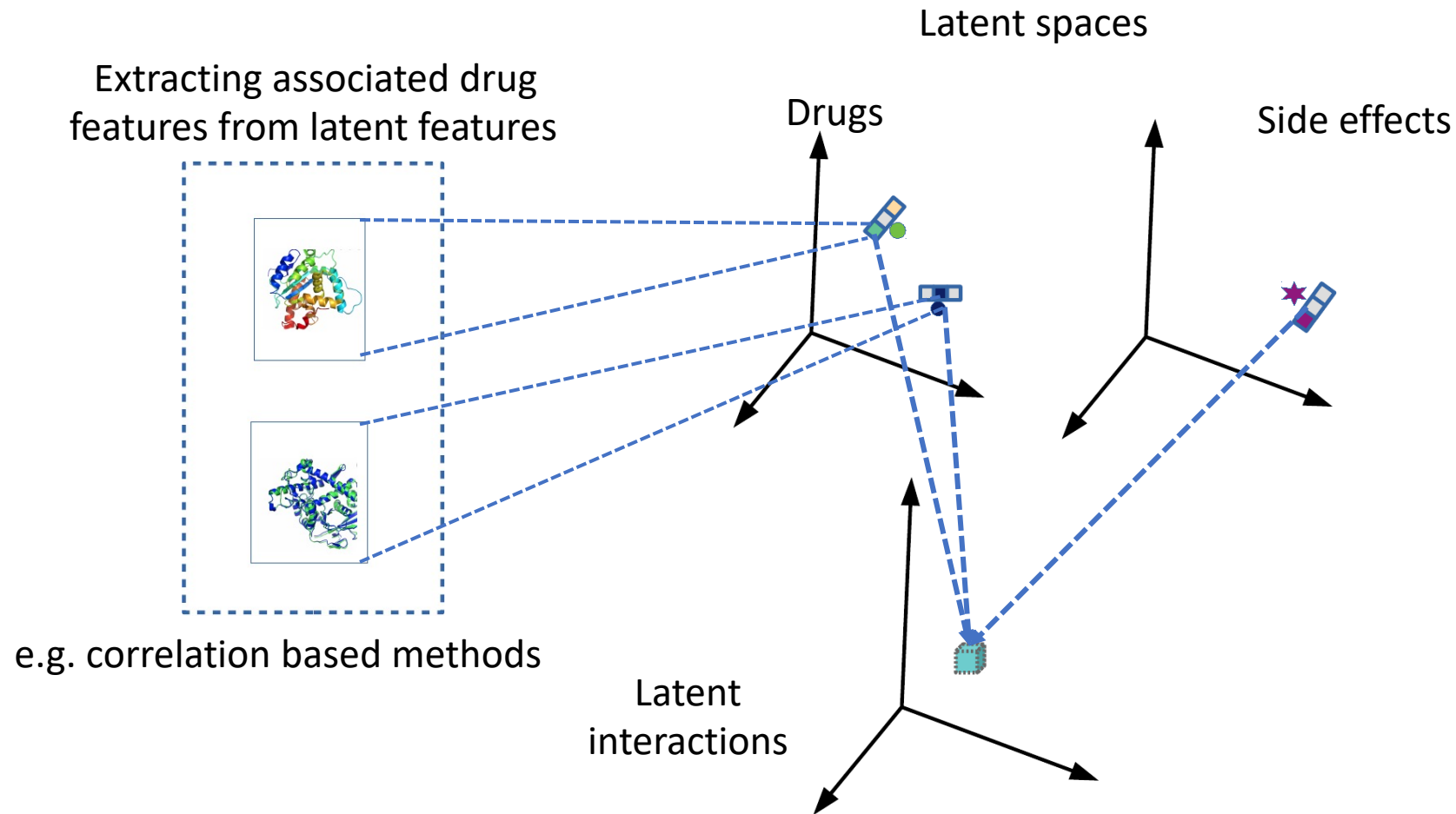
$$p(\Lambda_{i,j,k}) = \frac{2}{\pi} \frac{1}{1 + \Lambda_{i,j,k}^2}$$

Objective function: $\operatorname{argmax}_{B, H, \Lambda \geq 0} \underbrace{\log p(G|B, H)}_{\text{Likelihood}} + \underbrace{\log p(B|\Lambda)}_{\text{Prior}}$

Likelihood

Prior

Interpreting DDIs from latent spaces in SPARSE



Use extracted features
to explain latent interactions

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Experimental design

Datasets:

Name	No. of drugs	No. of side effects	No. of Drug-drug pairs	No. of Drug-drug-side effects (DDIs)	Sparsity (%)
TWOSIDES	557	964	48,677	3,606,046	97.6
CADDDI	587	969	21,918	373,976	99.77
JADERDDI	545	922	38,929	222,081	99.83

Performance measures:

- AUC (Area under the ROC curve)
- AUPR (Area under the precision-recall curve)

Train-test splitting:

- 20-fold cross-validation

Results for prediction performances

Method	TWO SIDES		CADD DI		JADERDDI	
	AUC	AUPR	AUC	AUPR	AUC	AUPR
MRGNN	0.8452 ± 0.0036	0.8029 ± 0.0039	0.9226 ± 0.0015	0.7113 ± 0.0031	0.9049 ± 0.0009	0.3698 ± 0.0019
Decagon	0.8639 ± 0.0029	0.8094 ± 0.0024	0.9132 ± 0.0014	0.6338 ± 0.0029	0.9099 ± 0.0012	0.4710 ± 0.0027
SpecConv	0.8785 ± 0.0025	0.8256 ± 0.0022	0.8971 ± 0.0055	0.6640 ± 0.0014	0.8862 ± 0.0025	0.5162 ± 0.0047
HPNN	0.9044 ± 0.0003	0.8410 ± 0.0007	0.9495 ± 0.0004	0.7020 ± 0.0018	0.9127 ± 0.0004	0.5198 ± 0.0016
SBM	0.9337 ± 0.0002	0.8583 ± 0.0004	0.9588 ± 0.0006	0.8170 ± 0.0008	0.9428 ± 0.0006	0.5963 ± 0.0018
CentSmoothie	0.9348 ± 0.0002	0.8749 ± 0.0013	0.9846 ± 0.0001	0.8230 ± 0.0019	0.9684 ± 0.0004	0.6044 ± 0.0025
SPARSE _O	0.9511 ± 0.0002	0.8811 ± 0.0001	0.9824 ± 0.0009	0.8773 ± 0.0014	0.9692 ± 0.0007	0.7230 ± 0.0008
SPARSE _L	0.9517 ± 0.0001	0.8815 ± 0.0002	0.9859 ± 0.0007	0.8797 ± 0.0010	0.9694 ± 0.0011	0.7276 ± 0.0017
SPARSE	0.9524 ± 0.0001	0.8820 ± 0.0002	0.9837 ± 0.0010	0.8843 ± 0.0012	0.9698 ± 0.0008	0.7348 ± 0.0018

Based lines: MRGNN: *Molecular graph NNs*. Decagon, SpecConv: *DDI graph NNs*. HPNN, SBM, CentSmoothie: *Existing DDI Hypergraph NNs*

SPARSE variants: SPARSE_O: No prior, SPARSE_L: Laplace prior, SPARSE: Horseshoe prior

- SPASRE without prior (SPARSE_O) is better than based lines in almost cases
-> Modeling multiple types of latent interactions is the main factor for high prediction performance.
- SPARSE with horseshoe prior achieves the highest performances
-> Horseshoe prior is suitable for SPARSE to control sparsity of latent interactions.

Samples for top predicted interactions with interpretations

No.	Drug A	Drug B	Side effect	Observable features associated with latent features	Extracted proteins of drugs from DrugBank	References
1	Ciprofloxacin	Mefenamic acid	Abdominal distension	Cytochrome enzymes	—	Venkataraman <i>et al.</i> (2014)
2	Naratriptan	Oxycodone	Abnormal ECG	Serotonin transporters and receptors	—	Baldo (2018) and Ritter <i>et al.</i> (2019)
3	Naratriptan	Tramadol	Abnormal ECG	Serotonin transporters and receptors	—	Baldo (2018)
4	Naratriptan	Sertraline	Abnormal ECG	Serotonin transporters and receptors	5-Hydroxytryptamine receptor 1B (and 1D) and sodium-dependent serotonin transporter	Ritter <i>et al.</i> (2019)
5	Naratriptan	Paroxetine	Abnormal ECG	Serotonin transporters and receptors	5-Hydroxytryptamine receptor 1B (and 1D) and sodium-dependent serotonin transporter	Ritter <i>et al.</i> (2019)
6	Trihexyphenidyl	Thiothixene	Abnormal EEG	Dopamine receptors	—	Ritter <i>et al.</i> (2019)
7	Carisoprodol	Orphenadrine	Abnormal vision	Not clear	—	Downs <i>et al.</i> (2019)
8	Buspirone	Orphenadrine	Abnormal vision	Not clear	—	Ritter <i>et al.</i> (2019)
9	Oxycodone	Orphenadrine	Abnormal vision	Not clear	—	Ritter <i>et al.</i> (2019)
10	Carisoprodol	Zaleplon	Abnormal vision	Not clear	—	Fagiolini <i>et al.</i> (2004)



Predicted DDI



Interpretation

SPARSE can extract associated drug features to interpret predicted DDI

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Conclusion

- We proposed SPARSE to encode DDIs in latent spaces using hypergraph neural networks and sparsity controls.
- SPARSE not only achieves high prediction performance but also can extract associated drug features for interpreting DDIs.

Thank you